

Balvinder Singh Nagi

Senior Software Engineering Manager | Enterprise Architect

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PROFESSIONAL PROFILE

Senior engineering leader and hands-on enterprise architect at Engineers India Limited since 2012. Built on-premise document platforms for EIL and subsidiary CEIL, with **estimated INR 10.5 crore gross product-cost avoidance** in a procurement, support and licensing model. Combines enterprise architecture, complete mobile product delivery, and on-premise AI.

Scenario: prior estimates + legacy charges + CEIL perpetual licences, setup and first-year support (50 assumed users). Internal costs excluded. Basis.

PROFESSIONAL EXPERIENCE

Engineers India Limited (EIL)

New Delhi, India

Engineering consultancy serving oil, gas, petrochemical, and infrastructure projects.

Senior Software Engineering Manager

Jul 2024 – Present

- As the department's **AI and Digital Transformation Coordinator**, introduced **EIL's first AI workflows in EngDMS**: document insights and queries, difference reporting, and AI-powered document reordering.
- Head **EIL's National Quantum Mission task force**, guiding the organization's participation and internal coordination.
- Architected and built **CEIL-eDMS**, a separate on-premise document platform for **EIL subsidiary CEIL**, from the ground up, with drawing review workflows, API rate limiting, IP-validated OTP, and brute-force defenses.
- Architected **AskData AI**, an on-premise Text-to-SQL assistant using Ollama, Next.js, and Oracle schema DDL parsing to support natural-language access to enterprise data.
- Built the **EIL Notification Hub**, consolidating legacy batch jobs into an event-driven pub/sub engine with pre- and post-dispatch hooks, department and role-based routing, and visual cron scheduling.
- Expanded the enterprise mobile app with field inspection and employee-service modules; modernized API integration and Oracle connection pooling and schema handling.

Software Engineering Manager

Jul 2020 – Jun 2024

- Built the **complete enterprise mobile app**, evolving a self-developed PWA into Flutter applications for **Android and iOS**, with employee services, HSE workflows, REST APIs, and **Firestore Cloud Messaging**.
- Secured management approval for **MacBook development hardware and app-store developer subscriptions**, enabling native build and release capability.
- Introduced **HAProxy** load balancing, TLS termination, and failover; managed RHEL/Hyper-V infrastructure and led migration of engineering records and workflows to the in-house EngDMS.
- Led delivery across five or more modules and mentored full-stack engineers; introduced Git branching standards, code review practices, and containerized deployment.

Deputy Software Engineering Manager

Jul 2016 – Jun 2020

- Received the **EIL Team Innovation Award** for the in-house **EngDMS™** platform, avoiding an estimated **INR 6 crore** in proprietary upgrade costs and reducing vendor dependency.
- Designed and launched the **Client Transmittal Portal**, digitizing inter-organizational drawing exchanges and avoiding an estimated **INR 1.5 crore** in vendor procurement costs.
- Built distributed document rendition workers; introduced automated **Let's Encrypt** issuance and renewal, now covering approximately **100 websites** and avoiding recurring certificate charges.

Senior Software Engineer / Engineers India Limited

Oct 2013 – Jun 2016

- Developed the **Vendor Document Management System**, digitizing vendor submissions and reducing document and drawing approval turnaround from **45 days to under 24 hours**.
- Built client and construction site portals with automated synchronization, reducing construction drawing issuance from **30 days to 1 day**.
- Deployed the IT department's first on-premise **GitLab Community Edition** through a Quality Circle initiative, now supporting approximately **100 developers**, with centralized version control, branch governance, and backups while avoiding paid seat licences.

Software Engineering Management Trainee / EIL

Oct 2012 – Sep 2013

- Developed foundational modules in Java, Spring MVC, and Oracle SQL while learning enterprise IT infrastructure and hydrocarbon engineering workflows.

SELECTED FREELANCE ENGINEERING PROJECTS**Real-time mobile applications & collaboration**

Flutter / Firebase / Supabase / AWS / GCP / Fastlane

Built mobile products with shared data, offline queues, push notifications, and location-based workflows. Implemented Android/iOS signing and release automation alongside messaging, task, and expense-tracking features.

AI-assisted publishing & commerce platforms

Next.js / React / Supabase / AWS / Razorpay

Developed resumable AI content generation, editorial tools, document export, credit ledgers, and payment integration. Built commerce interfaces with product administration and image-processing workflows.

Financial data pipelines & analytics

Python / Next.js / Ollama / GCP

Built ingestion and normalization pipelines for public financial disclosures, cached data APIs, and interactive dashboards. Combined rules-based extraction with source-grounded local-model summaries.

TECHNICAL EXPERTISE

Leadership Enterprise architecture, end-to-end delivery, mentoring, cross-functional collaboration, code review, and engineering governance.

Backend & data Java, Spring Boot, Spring MVC, Node.js, Express, REST APIs, ActiveMQ/JMS, Oracle, PL/SQL, MongoDB, Redis, Supabase.

AI & applications Python, Ollama, Text-to-SQL, prompt engineering, PyTorch, React, Next.js, TypeScript, JavaScript, Flutter, Dart, Firebase/FCM.

Cloud platforms AWS, Google Cloud Platform (GCP), Firebase, and Supabase.

Infrastructure & delivery HAProxy, RHEL, Hyper-V, Active Directory, Docker, Git, GitLab CE, CI/CD, Fastlane, TLS automation, Playwright.

EDUCATION & PROFESSIONAL DEVELOPMENT**AI Programming with Python Nanodegree**

Nov 2022 – Mar 2023

Udacity / PyTorch, computer vision, and linear algebra.

B.Tech., Computer Science & Engineering

2008 – 2012

Guru Gobind Singh Indraprastha University, Delhi.

Additional coursework: Oracle (HP Education Services, 2011); .NET Framework (HP Education Services, 2010); C and C++ (APTECH, 2008–2009).